



RevibeFit - Domain-Driven Design Document

1. Title and Project Description

Title: RevibeFit - A Comprehensive Fitness and Health Ecosystem Platform

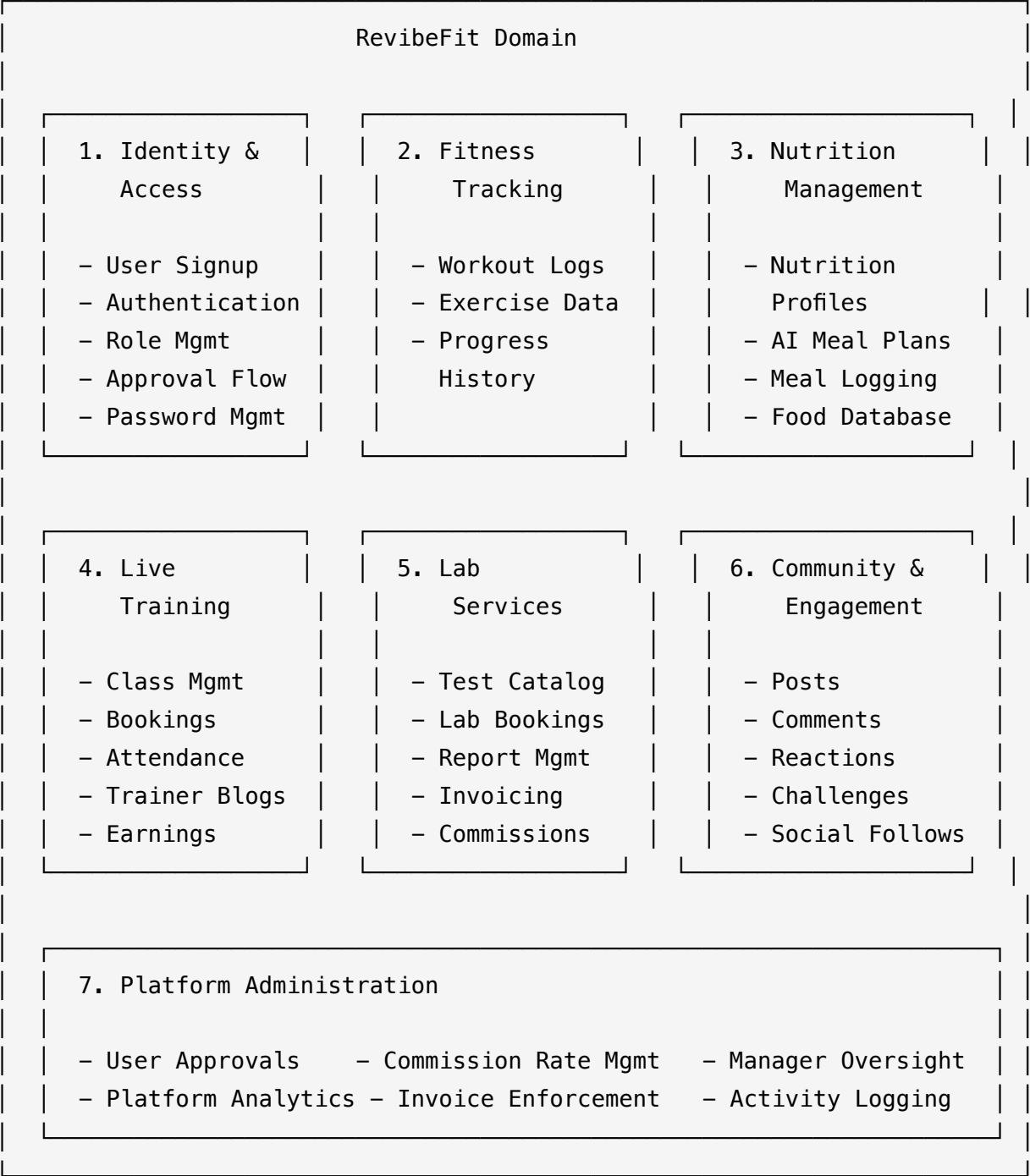
Description

RevibeFit is a full-stack web platform that connects fitness enthusiasts, certified trainers, diagnostic lab partners, and platform administrators within a unified health and wellness ecosystem. The platform empowers fitness enthusiasts to track workouts, receive AI-powered personalized meal plans (via Google Gemini), book live training classes, access diagnostic lab services, and engage with a fitness-focused community. Trainers can conduct live classes, publish educational blogs, manage their client base, and track earnings with commission-based payouts. Lab partners manage diagnostic test catalogs, process bookings, upload digital reports, and handle commission-based invoicing. Administrators oversee user approvals, platform analytics, commission rate management, invoice generation, and regional manager assignments. Managers provide regional oversight of trainers and lab partners, assisting with approvals and monitoring activity. The platform features role-based access control, real-time communication via [Socket.io](#), a multi-stage financial workflow (commission tracking, invoice generation, grace-period enforcement, and automated suspension for non-payment), and a community module with posts, challenges, reactions, and threaded discussions. Built on the MERN stack (MongoDB, Express.js, React 18, Node.js), RevibeFit delivers a production-grade, multi-stakeholder health platform with enterprise-level financial controls and AI-driven personalization. (196 words)

2. Domain-Driven Design

2a. Bounded Contexts

The system is decomposed into **seven** bounded contexts, each encapsulating a cohesive domain area with its own ubiquitous language, models, and rules.

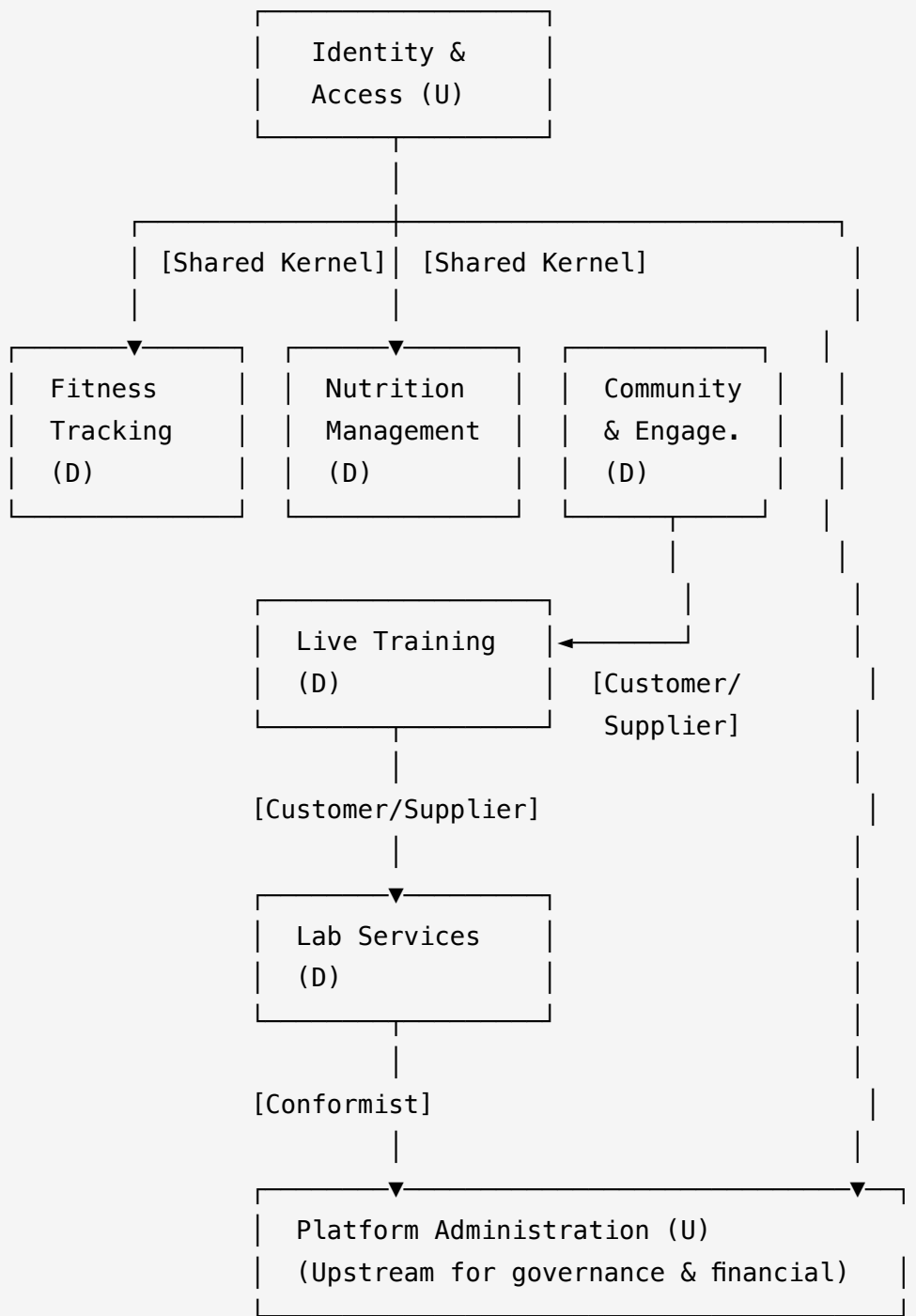


#	Bounded Context	Responsibility
1	Identity & Access	User registration, authentication (JWT), role-based authorization, approval workflows, password management

#	Bounded Context	Responsibility
2	Fitness Tracking	Logging completed workouts, tracking exercise history, fitness progress
3	Nutrition Management	Nutrition profiles, AI-generated meal plans (Gemini), meal logging, food search (FatSecret API), nutrition analytics
4	Live Training	Live class creation/scheduling by trainers, class bookings, attendance, refunds, trainer earnings, blog publishing
5	Lab Services	Diagnostic test catalog, lab bookings, report upload/download, commission billing, platform invoicing
6	Community & Engagement	Community posts, threaded comments, reactions, fitness challenges, participant tracking, social follows
7	Platform Administration	Admin approval/rejection, analytics dashboards, commission rate management, invoice enforcement, manager creation, activity audit logs

2b. Context Mappings

Context mappings define the relationships and integration patterns between bounded contexts.



Upstream (U)	Downstream (D)	Mapping Pattern	Description
Identity & Access	Fitness Tracking	Shared Kernel	Both share the User entity; fitness tracking depends on authenticated

Upstream (U)	Downstream (D)	Mapping Pattern	Description
			user identity
Identity & Access	Nutrition Management	Shared Kernel	Nutrition profiles are linked to authenticated users via <code>userId</code>
Identity & Access	Live Training	Shared Kernel	Trainers and fitness enthusiasts are User subtypes; class bookings reference user identity
Identity & Access	Lab Services	Shared Kernel	Lab partners and fitness enthusiasts are User subtypes; bookings reference both
Identity & Access	Community & Engagement	Shared Kernel	Posts, comments, follows, and challenges reference user identity
Identity & Access	Platform Administration	Shared Kernel	Admin and Manager are User subtypes; approval workflows modify user state
Live Training	Community & Engagement	Customer/ Supplier	Trainers (from Live Training) create challenges and blog content consumed by the community
Lab Services	Platform Administration	Conformist	Lab Services conforms to invoice structures and commission rules defined by administration
Live Training	Platform Administration	Conformist	Trainer earnings and commission rates are governed by admin-defined policies
Platform Administration	Lab Services	Customer/ Supplier	Admin generates invoices, sets commission rates; Lab Services consumes these policies
Platform	Live Training	Customer/	Admin manages trainer commission

Upstream (U)	Downstream (D)	Mapping Pattern	Description
Administration		Supplier	rates and earnings oversight
Nutrition Management	Fitness Tracking	Separate Ways	Both operate independently; no direct dependency (both relate only to User)

2c. Entities, Value Objects, and Services

Bounded Context 1: Identity & Access

Type	Name	Description
Entity	User	Central identity entity; polymorphic by <code>userType</code> (fitness-enthusiast, trainer, lab-partner, admin, manager)
Value Object	Credentials	Email + hashed password combination
Value Object	SocialLinks	Trainer's social media URLs (Instagram, YouTube, Twitter, LinkedIn, Website)
Value Object	OperatingHours	Lab partner's daily operating schedule (open/close times per day)
Value Object	ApprovalStatus	Status enum (pending, approved, rejected) with approval metadata
Value Object	RefreshToken	JWT refresh token for session management
Value Object	PasswordResetToken	Token + expiry for password recovery
Service	AuthService	Signup, login, logout, JWT generation/validation, cookie management

Type	Name	Description
Service	PasswordService	Password hashing (bcrypt), change password, forgot/reset password via email
Service	EmailService	Nodemailer-based transactional email (reset links, approval notifications)

Bounded Context 2: Fitness Tracking

Type	Name	Description
Entity	CompletedWorkout	Record of a completed workout session with duration, difficulty, and category
Value Object	Difficulty	Enum: Beginner, Intermediate, Advanced
Value Object	WorkoutCategory	Category classification string
Service	WorkoutService	Log completed workouts, retrieve history, delete entries

Bounded Context 3: Nutrition Management

Type	Name	Description
Entity	NutritionProfile	User's dietary profile with auto-calculated metrics (BMI, BMR, TDEE, daily targets)
Entity	MealPlan	AI-generated weekly meal plan with detailed nutritional breakdown per meal
Entity	MealLog	Daily meal log entries with food items and nutrition data
Value Object	FoodItem	Individual food entry with calories, protein, carbs, fats, fiber, sugar, sodium

Type	Name	Description
Value Object	DailyMeal	Single day's meal structure (breakfast, lunch, dinner, snacks) within a MealPlan
Value Object	MacroTargets	Calculated daily targets: calories, protein, carbs, fats
Value Object	BodyMetrics	Height, weight, BMI, BMR, TDEE - auto-calculated from profile data
Value Object	DietaryPreference	Enum: vegetarian, vegan, keto, paleo, mediterranean, etc.
Value Object	ActivityLevel	Enum: sedentary, lightly-active, moderately-active, very-active, extremely-active
Value Object	FitnessGoal	Enum: weight-loss, muscle-gain, maintenance, endurance, general-health
Service	NutritionProfileService	CRUD for nutrition profiles; auto-calculates BMI/BMR/TDEE/macro targets
Service	MealPlanService	AI meal plan generation (Google Gemini), CRUD for meal plans
Service	MealLogService	Log meals, retrieve by date range, daily summaries, nutrition statistics
Service	FoodSearchService	FatSecret API integration for food database search and nutritional analysis

Bounded Context 4: Live Training

Type	Name	Description
Entity	LiveClass	Scheduled training session with class type, capacity, cost, and status
Entity	ClassBooking	User's booking for a live class with payment, attendance, and refund tracking

Type	Name	Description
Entity	Blog	Educational content published by trainers with categories and thumbnails
Entity	BlogReading	Tracks which user has read which blog (unique per user-blog pair)
Value Object	ClassType	Enum: cycling, strength, running, yoga, meditation, rowing, outdoor, stretching, other
Value Object	BookingStatus	Enum: active, cancelled, completed, no-show
Value Object	PaymentStatus	Enum: pending, completed, failed, refunded
Value Object	AttendanceStatus	Enum: registered, attended, missed
Value Object	RefundPolicy	Time-based refund: >24hrs = 100%, 2-24hrs = 50%, <2hrs = 0%
Value Object	CommissionDetails	Rate, amount, trainer payout - calculated per booking
Value Object	BlogCategory	Enum: Fitness Tips, Nutrition, Yoga, Mental Wellness, General
Service	LiveClassService	Class creation/update by trainers, status auto-transitions, capacity management
Service	ClassBookingService	Join/cancel classes, refund calculation, booking history, attendance tracking
Service	BlogService	Blog CRUD, publish/unpublish, reading tracking
Service	TrainerDashboardService	Dashboard analytics, client listing, schedule view, earnings calculation

Bounded Context 5: Lab Services

Type	Name	Description
Entity	LabTest	Diagnostic test offered by a lab partner with price, duration, and category
Entity	LabBooking	Booking of one or more lab tests by a fitness enthusiast
Entity	PlatformInvoice	Invoice issued to lab partners for accumulated commissions over a billing period
Value Object	TestCategory	Enum: Blood Test, Urine Test, Imaging, Fitness Assessment, Cardiac Test, Other
Value Object	SelectedTest	Embedded test reference with snapshot of testName and price at booking time
Value Object	BookingStatus	Enum: pending, confirmed, completed, cancelled
Value Object	CommissionStatus	Enum: pending, billed, paid
Value Object	BillingPeriod	Month/year or custom date range for invoice generation
Value Object	CommissionBreakdown	Per-booking details within an invoice (booking ID, test names, amounts, rates)
Value Object	InvoiceStatus	Enum: payment_due, paid, overdue, cancelled
Value Object	UserPaymentInfo	Payment method (cash, card, online, UPI), date, verified-by
Service	LabTestService	Test catalog CRUD, offered tests management
Service	LabBookingService	Create/cancel bookings, status transitions, report upload/download

Type	Name	Description
Service	LabPaymentService	Mark user payment received, commission calculation, payment verification
Service	InvoiceService	Generate monthly/custom invoices, mark paid, financial summary

Bounded Context 6: Community & Engagement

Type	Name	Description
Entity	CommunityPost	User-authored post with content, images, tags, and engagement metrics
Entity	Comment	Threaded comment on a post (supports nested replies via self-reference)
Entity	Reaction	User reaction on a post or comment (polymorphic target)
Entity	Challenge	Fitness challenge created by trainers/admins with goals, rules, and duration
Entity	ChallengeParticipant	User's participation record with progress logs, completion status, and rank
Entity	Follow	Social follow relationship between users (many-to-many)
Value Object	PostCategory	Enum: tip, question, transformation, motivation, discussion, success-story
Value Object	ReactionType	Enum: like, love, fire, clap
Value Object	GoalType	Enum: count, duration, streak
Value Object	ProgressLogEntry	Timestamped progress entry with value and optional note

Type	Name	Description
Value Object	ChallengeDifficulty	Enum: beginner, intermediate, advanced
Service	CommunityPostService	Post CRUD, pinning, visibility control, image uploads
Service	CommentService	Threaded comment CRUD, visibility management
Service	ReactionService	Add/remove reactions on posts and comments
Service	ChallengeService	Challenge CRUD, participant management, progress tracking, leaderboards
Service	FollowService	Follow/unfollow users, follower/following lists

Bounded Context 7: Platform Administration

Type	Name	Description
Entity	ManagerActivityLog	Audit trail of manager actions with target, details, and IP address
Entity	CommissionChangeRequest	Request to modify commission rates (by managers), with admin approval flow
Value Object	ManagerType	Enum: trainer_manager, lab_manager
Value Object	ManagerAction	Enum of 18 tracked actions (APPROVE_USER, GENERATE_INVOICE, etc.)
Value Object	AssignedRegions	Array of Indian geographic regions assigned to a manager
Value Object	RequestStatus	Enum: pending, approved, denied
Service	ApprovalService	Pending approvals, approve/reject users,

Type	Name	Description
		request additional info, claim/release locking
Service	PlatformAnalyticsService	User stats, monthly growth, user distribution, dashboard analytics
Service	CommissionManagementService	View/update commission rates for trainers and lab partners (5-30% range)
Service	InvoiceEnforcementService	Generate invoices, enforce overdue payments, suspend/unsuspend labs, grace period management
Service	ManagerService	Manager CRUD, region assignment, activity log retrieval, soft/hard delete



2d. Cardinality Ratios

CARDINALITY RATIO DIAGRAM

User (1)	—————	(0..N)	CompletedWorkout	
User (1)	—————	(0..1)	NutritionProfile	
User (1)	—————	(0..N)	MealPlan	
User (1)	—————	(0..N)	MealLog	
User (1)	—————	(0..N)	LiveClass	[as Trainer]
User (1)	—————	(0..N)	ClassBooking	[as Enthusiast]
User (1)	—————	(0..N)	Blog	[as Trainer]
User (1)	—————	(0..N)	BlogReading	[as Enthusiast]
User (1)	—————	(0..N)	LabTest	[as Lab Partner]
User (1)	—————	(0..N)	LabBooking	[as Enthusiast]
User (1)	—————	(0..N)	LabBooking	[as Lab Partner]
User (1)	—————	(0..N)	PlatformInvoice	[as Lab Partner]
User (1)	—————	(0..N)	CommunityPost	
User (1)	—————	(0..N)	Comment	
User (1)	—————	(0..N)	Reaction	
User (1)	—————	(0..N)	Challenge	[as Creator]
User (1)	—————	(0..N)	ChallengeParticipant	
User (1)	—————	(0..N)	ManagerActivityLog	[as Manager]
User (1)	—————	(0..N)	CommissionChangeRequest	
User (1)	—————	(0..N)	Follow	[as Follower]
User (1)	—————	(0..N)	Follow	[as Following]
User (1)	—————	(0..N)	User	[reportsTo]
LiveClass (1)	—————	(0..N)	ClassBooking	
Blog (1)	—————	(0..N)	BlogReading	
CommunityPost (1)	—————	(0..N)	Comment	
Comment (1)	—————	(0..N)	Comment	[Threaded]
Challenge (1)	—————	(0..N)	ChallengeParticipant	
LabBooking (N)	—————	(1)	PlatformInvoice	
LabBooking (N)	—————	(M)	LabTest	[Embedded]
User (M)	—————	(N)	User	[via Follow]
User (1)	—————	(1)	ClassBooking	[per LiveClass]
User (1)	—————	(1)	ChallengeParticipant	[per Challenge]
User (1)	—————	(1)	BlogReading	[per Blog]
Reaction (N)	—————	(1)	CommunityPost OR Comment	[Polymorphic]

Entity A	Cardinality	Entity B	Constraint
User	1 : N	CompletedWorkout	One user logs many workouts
User	1 : 1	NutritionProfile	Exactly one profile per user (unique index)
User	1 : N	MealPlan	One user has many meal plans
User	1 : N	MealLog	One user logs many meals
User (Trainer)	1 : N	LiveClass	One trainer creates many classes
User (Enthusiast)	1 : N	ClassBooking	One user books many classes
LiveClass	1 : N	ClassBooking	One class has many bookings
User + LiveClass	1 : 1	ClassBooking	Unique: one booking per user per class
User (Trainer)	1 : N	Blog	One trainer writes many blogs
User	1 : N	BlogReading	One user reads many blogs
User + Blog	1 : 1	BlogReading	Unique: one reading per user per blog
Blog	1 : N	BlogReading	One blog read by many users
User (Lab Partner)	1 : N	LabTest	One lab offers many tests

Entity A	Cardinality	Entity B	Constraint
User (Enthusiast)	1 : N	LabBooking	One user makes many lab bookings
User (Lab Partner)	1 : N	LabBooking	One lab receives many bookings
LabBooking	N : M	LabTest	Many tests selected per booking (embedded)
User (Lab Partner)	1 : N	PlatformInvoice	One lab receives many invoices
PlatformInvoice	1 : N	LabBooking	One invoice covers many bookings
User	1 : N	CommunityPost	One user creates many posts
CommunityPost	1 : N	Comment	One post has many comments
Comment	1 : N	Comment	Parent-child threaded replies
User	1 : N	Comment	One user writes many comments
User	1 : N	Reaction	One user makes many reactions
CommunityPost / Comment	1 : N	Reaction	One target receives many reactions (polymorphic)
User + Target	1 : 1	Reaction	Unique: one reaction per user per target
User (Trainer/	1 : N	Challenge	One creator makes

Entity A	Cardinality	Entity B	Constraint
Admin)			many challenges
Challenge	1 : N	ChallengeParticipant	One challenge has many participants
User	1 : N	ChallengeParticipant	One user joins many challenges
User + Challenge	1 : 1	ChallengeParticipant	Unique: one participation per user per challenge
User	M : N	User	Many-to-many via Follow (follower ↔ following)
User (Manager)	1 : N	ManagerActivityLog	One manager has many activity log entries
User	1 : N	CommissionChangeRequest	One user has many commission requests
User (Manager)	N : 1	User (Manager)	Manager hierarchy via reportsTo

2e. Aggregates

Aggregates define consistency boundaries - the **Aggregate Root** controls access to all entities within the aggregate.

AGGREGATE 1: User Identity Aggregate

Root: User

User (Root)	— Credentials (V0)
	— ApprovalStatus (V0)
	— SocialLinks (V0)
	— OperatingHours (V0)
	— RefreshToken (V0)
	— PasswordResetToken (V0)

Invariant: Email must be unique; password always hashed

Invariant: Trainer/Lab approval required before activation

AGGREGATE 2: Workout Aggregate

Root: CompletedWorkout

CompletedWorkout (Root)	— Difficulty (V0)
	— WorkoutCategory (V0)

Invariant: Each workout belongs to exactly one user

AGGREGATE 3: Nutrition Profile Aggregate

Root: NutritionProfile

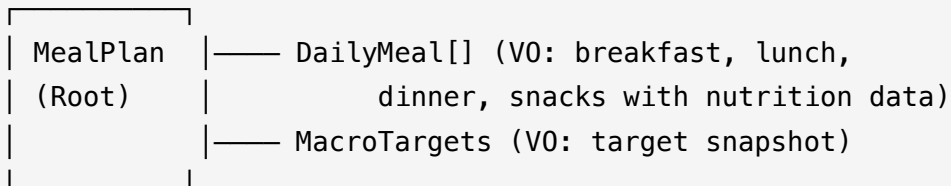
NutritionProfile (Root)	— BodyMetrics (V0: BMI, BMR, TDEE)
	— MacroTargets (V0)
	— FitnessGoal (V0)
	— DietaryPreference (V0)
	— ActivityLevel (V0)

Invariant: One profile per user (unique constraint)

Invariant: BMI/BMR/TDEE auto-recalculated on profile update

AGGREGATE 4: Meal Plan Aggregate

Root: MealPlan

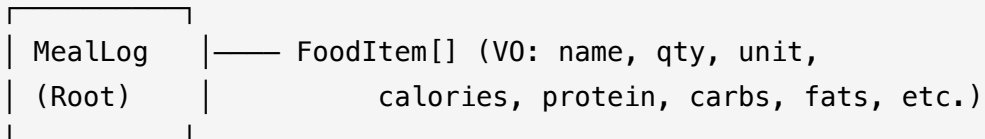


Invariant: Daily totals auto-calculated from meal items

Invariant: Only one active meal plan per user at a time

AGGREGATE 5: Meal Log Aggregate

Root: MealLog

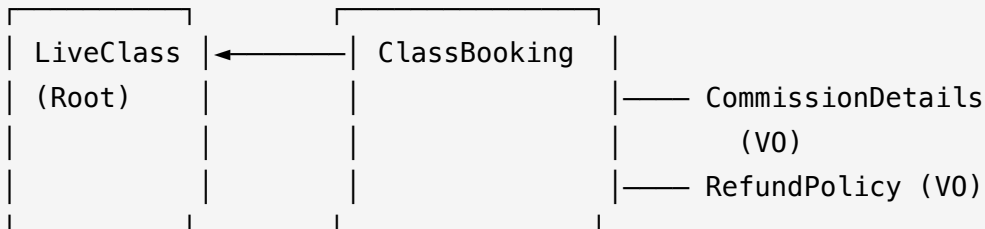


Invariant: At least one food item per meal log

Invariant: Totals computed virtually from food items

AGGREGATE 6: Live Class Aggregate

Root: LiveClass



Invariant: $\text{currentParticipants} \leq \text{maxParticipants}$

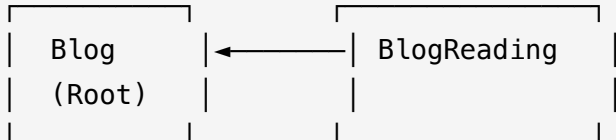
Invariant: One booking per user per class

Invariant: Status transitions: scheduled→ongoing→completed

Invariant: Refund calculated based on time before class start

AGGREGATE 7: Blog Aggregate

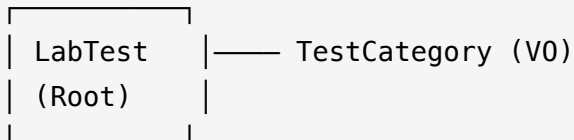
Root: Blog



Invariant: One BlogReading per user per blog

AGGREGATE 8: Lab Test Catalog Aggregate

Root: LabTest

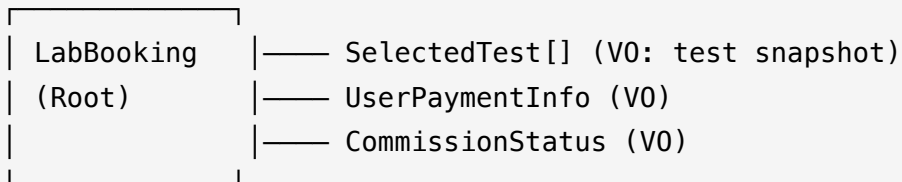


Invariant: Test belongs to exactly one lab partner

Invariant: Price ≥ 0

AGGREGATE 9: Lab Booking Aggregate

Root: LabBooking



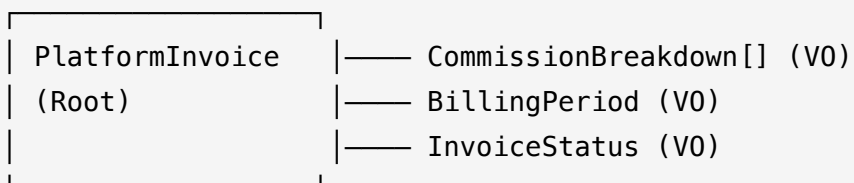
Invariant: totalAmount = sum of selected test prices

Invariant: Commission calculated only after user payment

Invariant: Status flow: pending→confirmed→completed

AGGREGATE 10: Platform Invoice Aggregate

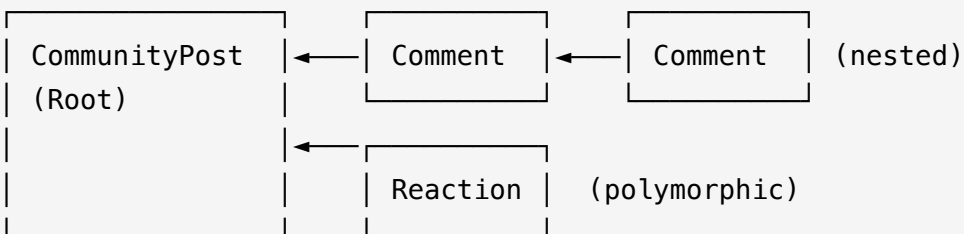
Root: PlatformInvoice



Invariant: Invoice number is unique and auto-generated
Invariant: Status flow: payment_due→paid (or →overdue)
Invariant: Overdue triggers lab suspension after grace period

AGGREGATE 11: Community Post Aggregate

Root: CommunityPost



Invariant: likesCount/commentsCount stay consistent
Invariant: One reaction per user per target

AGGREGATE 12: Challenge Aggregate

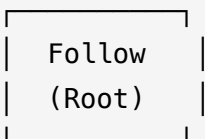
Root: Challenge



Invariant: participantsCount ≤ maxParticipants
Invariant: One participation per user per challenge
Invariant: Progress only logged between startDate and endDate

AGGREGATE 13: Follow Aggregate

Root: Follow



Invariant: Unique pair (follower, following)
Invariant: User cannot follow themselves

AGGREGATE 14: Manager Activity Aggregate

Root: ManagerActivityLog

ManagerActivityLog (Root) — ManagerAction (V0)
— ManagerType (V0)

Invariant: Every manager action is logged (append-only)

AGGREGATE 15: Commission Change Request Aggregate

Root: CommissionChangeRequest

CommissionChangeRequest (Root) — RequestStatus (V0)

Invariant: Proposed rate must be between 0-100%

Invariant: Only admin can approve/deny

Aggregate Summary Table

#	Aggregate Root	Contained Entities	Key Value Objects	Bound Context
1	User	—	Credentials, ApprovalStatus, SocialLinks, OperatingHours, RefreshToken	Identity & Access
2	CompletedWorkout	—	Difficulty, WorkoutCategory	Fitness Tracking
3	NutritionProfile	—	BodyMetrics, MacroTargets,	Nutrition Managem

#	Aggregate Root	Contained Entities	Key Value Objects	Boundary Context
			FitnessGoal, DietaryPreference, ActivityLevel	
4	MealPlan	—	DailyMeal[], MacroTargets	Nutrition Management
5	MealLog	—	FoodItem[]	Nutrition Management
6	LiveClass	ClassBooking	CommissionDetails, RefundPolicy, ClassType, BookingStatus	Live Train
7	Blog	BlogReading	BlogCategory	Live Train
8	LabTest	—	TestCategory	Lab Servi
9	LabBooking	—	SelectedTest[], UserPaymentInfo, CommissionStatus	Lab Servi
10	PlatformInvoice	—	CommissionBreakdown[], BillingPeriod, InvoiceStatus	Lab Servi
11	CommunityPost	Comment, Reaction	PostCategory, ReactionType	Communi Engagem
12	Challenge	ChallengeParticipant	GoalType, ProgressLogEntry[], ChallengeDifficulty	Communi Engagem
13	Follow	—	—	Communi Engagem
14	ManagerActivityLog	—	ManagerAction,	Platform

#	Aggregate Root	Contained Entities	Key Value Objects	Bound Conte
			ManagerType	Administr
15	CommissionChangeRequest	—	RequestStatus	Platform Administr